

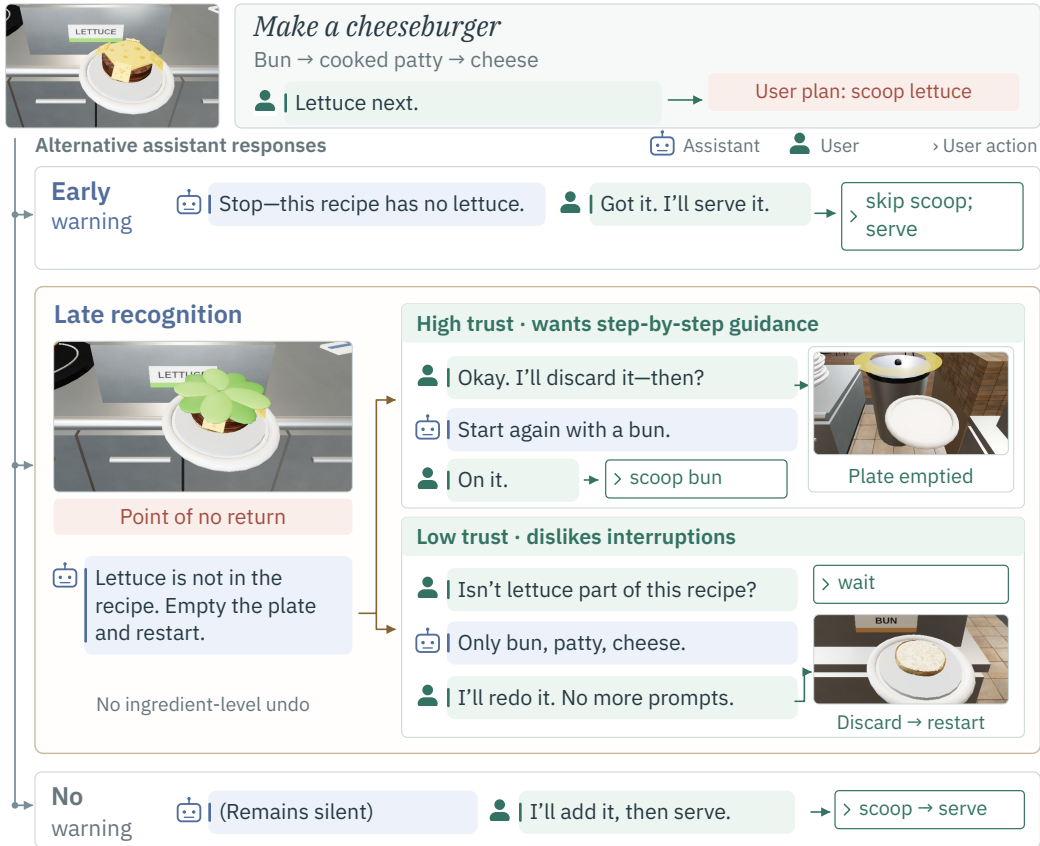


Figure 1: **Closed-loop interaction with simulated users.** Early intervention prevents an off-recipe addition; late recognition prompts recovery, while silence leaves the error unaddressed. The late case shows how user personas change the conversation and subsequent actions. Authored dialogue illustrates engine-executed trajectories. Once an ingredient is plated, this simulator requires emptying the whole plate to recover.

in AI, and interruption preference. These attributes shape how real-world users perform tasks and respond to guidance differently, enabling evaluation of the assistant’s adaptation to different users.

Third, simulation provides execution traces that support comprehensive evaluation of an assistant, covering task completion, interaction efficiency, error detection and suggestion quality, and overall interaction quality. This captures how assistance affects task execution and user experience, which offline replay cannot assess.

As an initial instantiation of our framework, we construct a benchmark with 150 cooking, 75 household, and 90 mobile-interface interaction tasks to evaluate state-of-the-art VLMs as visual assistants. In each episode, the assistant is tasked to support a simulated user with a sampled persona through task execution, and needs to deal with a set of scheduled events, including execution errors, mid-task plan changes, questions, and requests for proactive assistance. [results]

[RL]

2 RELATED WORK

Streaming video understanding and proactive assistance. Egocentric procedural datasets support step recognition, mistake detection and assistance during ongoing activities (Wang et al., 2023; Peddi et al., 2024; Flaborea et al., 2024), and recent benchmarks study when a model should respond, including correct silence and early versus late responses (Lin et al., 2024; Ran et al., 2026; Zhang et al., 2025; Liu et al., 2026; Kundu et al., 2026; Bhattacharyya et al., 2026; Azad et al., 2026). Their evaluation remains open-loop: the video and its outcome are fixed before the model responds.

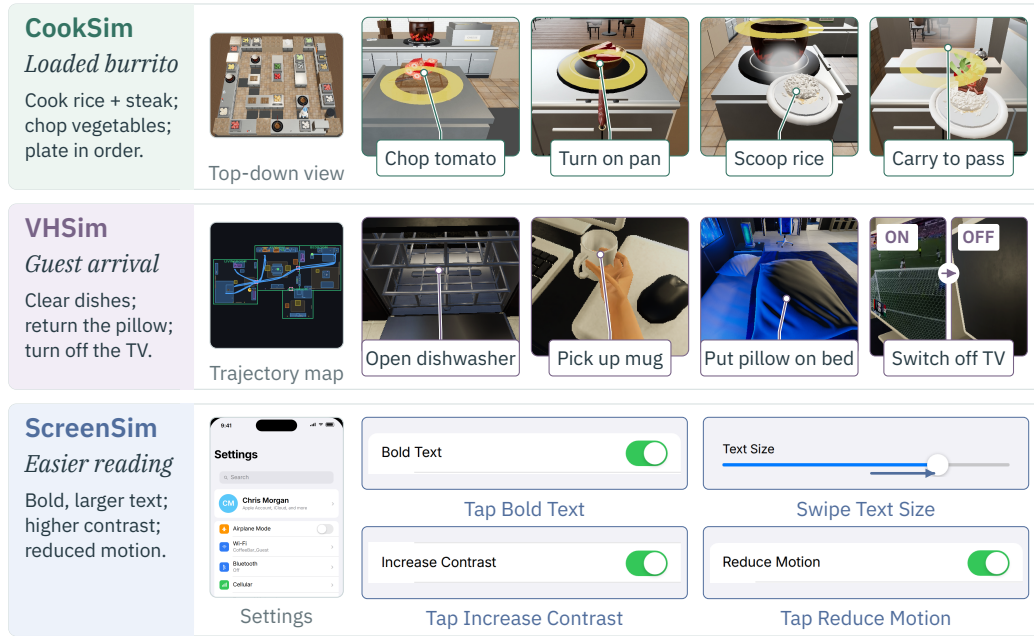


Figure 2: **Example tasks and observations.** Selected execution checkpoints illustrate preparing a loaded burrito, preparing an apartment for guests, and improving phone readability. Callouts identify action targets; television views compare before/after states, and phone panels magnify native controls from different screens.

	Task	Goal	Action classes
CookSim	loaded_burrito	Cook rice and sliced steak; chop tomato and lettuce; plate in that order.	<i>Navigate:</i> go to <food source>; go to <cook station>;
	shrimp_pasta	Boil noodles, chop tomato, and sear shrimp; plate in that order.	go to <prep station>; go to <counter>; go to plates/pass/trash.
	steak_frites	Sear steak slices; fry two chip portions; plate steak first.	<i>Manipulate:</i> pick up; put down; scoop; chop; turn on; trash; serve.
VHSim	post_meal_cleanup	Load dishes into the dishwasher; return the apple to the fridge; close both.	<i>Navigate:</i> turn left/right; go straight; look up/down; go through the visible door; turn to <obj>; go to <visible obj/place>.
	collect_dishes	Gather three plates and two glasses; load and close the dishwasher.	<i>Manipulate:</i> pick up <obj>; put <obj> on/in <target>; open/close <obj>; switch on/off <obj>; drink <obj>; sit on <obj>; stand up.
	return_belongings	Return toilet paper to the bathroom counter, a book to the desk, and a pillow to the bed.	
ScreenSim	app_privacy	Disable one app’s location, tracking, and background refresh only.	<i>Input:</i> tap <label>; type <text>.
	readability	Enlarge and bold text; clarify buttons; increase contrast; reduce motion.	<i>Navigate and adjust:</i> swipe up/down; swipe left/right on <label>; press back/home.
	lock_screen_security	Block Control Center, Today View, Wallet, and missed-call return.	

Table 1: **Environment overview.** Example tasks and action classes.

Progress tracking via scene-graph state. Each simulator represents world state as a scene graph whose nodes encode objects and attributes, such as a dishwasher’s open or closed state, and whose edges encode containment and support relations. We decompose each task into subgoals defined by conditions over partial graph state, such as the designated dishes being inside the dishwasher. Tracking these conditions measures task progress independently of different valid execution procedures.